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Kasko Lab & Hydrogel Wound Dressing Project

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Kasko Lab

NOVEL POLYMERIC BIOMATERIALS

"*Polymer*"- large, chain-like molecule made up of monomer repeat units, which are small molecules

- Novel antimicrobials
- Tobacco-DNA interactions
- Sickle cell therapy
- Plant-based polymers
- Gene delivery

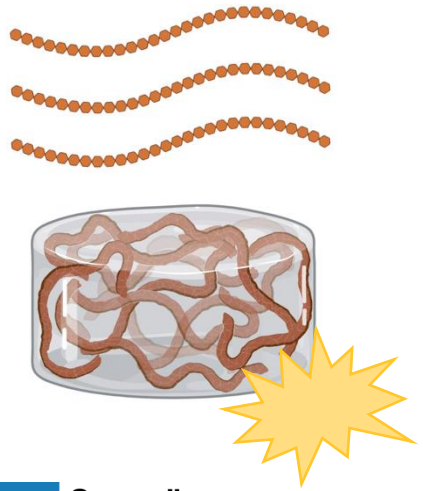


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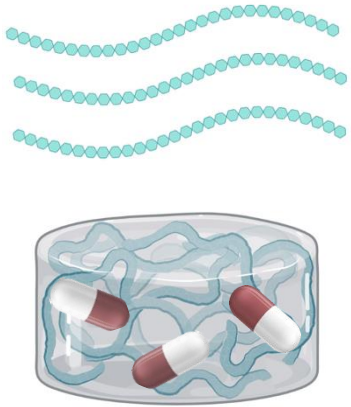
Hydrogel Projects

"*Hydrogel*"- crosslinked 3D network of hydrophilic polymers that swells in water, while maintaining its structure

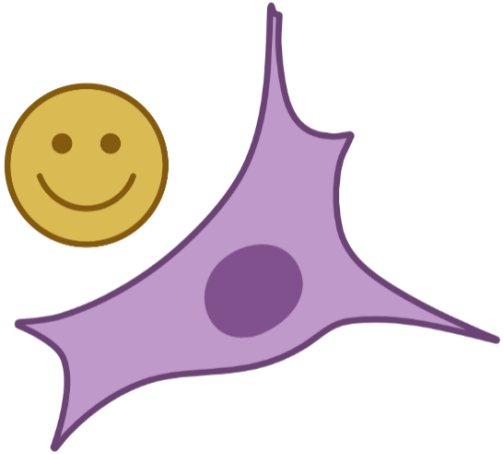
PHOTORESPONSIVE GELS



CHITOSAN GELS



HYDROGEL INITIATOR SYSTEMS



DoD Hydrogel Project

DRUG-LOADED HYDROGEL WOUND DRESSING

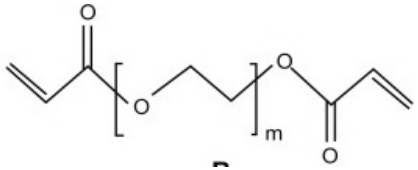
Goal: develop a field-polymerizable hydrogel wound dressing—designed for use in harsh environments—capable of release of antibiotics, analgesics (pain relief), and a haemostatic agent (stops bleeding) over a 3- to 5-day period

IDEAL SYSTEM:

- Powder → gel in less than 60 seconds
- No additional equipment (mobile)
- Able to withstand harsh environments
- Easy to add or combine different drugs

Hydrogel Components

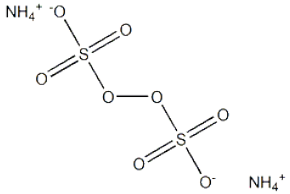
1. Polymer

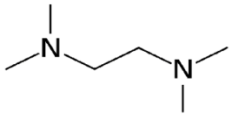


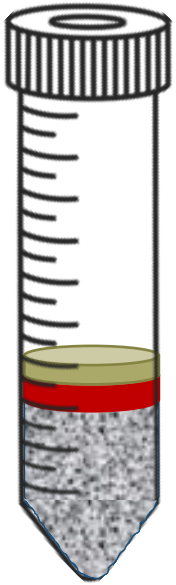
2. Therapeutic agent



3. Initiators







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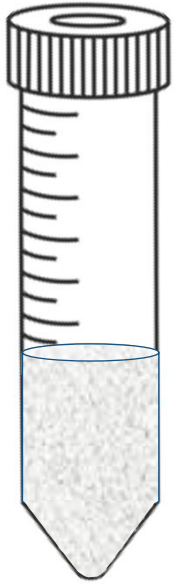
Hydrogel Components



1. Polymer

2. Therapeutic agent

3. Initiators

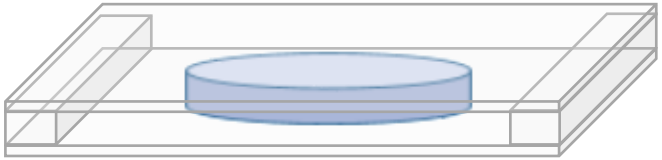


After mixing, hydrogel will be poured into wound, solidifying within 60 s to provide sustained drug delivery

Quantifying Therapeutic Release Rate

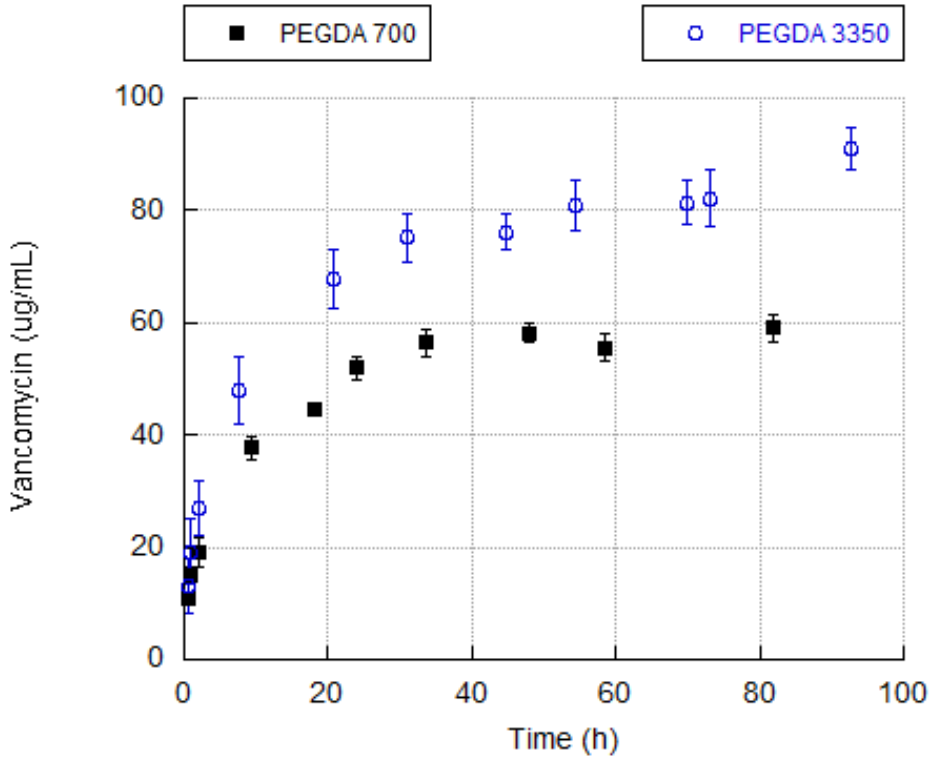
TEST SYSTEM IN VITRO ON SMALLER SCALE:

- Cast gels onto glass slide with 2 mm spacers, cut to standard size
- Place each gel sample in known volume of DI H_2O in well plate
- Take aliquot every ~8 hours, measure using UV Vis (measures absorbance of light)
- Absorbance relates to amount of drug that has been released from hydrogel sample





Example Release Profile



PEGDA: polymer

700: low molecular weight

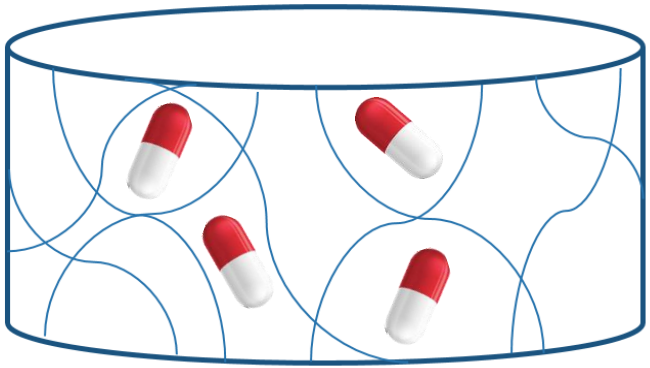
3350: high molecular weight

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Tailoring Therapeutic Release Rate

Drug release from hydrogels is **diffusion-dependent**, which is affected by:

- Polymer molecular weight
- Hydrogel mesh size
- Drug concentration
- Drug molecular weight



Polymer molecular weight changes the gel's mesh size

Denser mesh makes it harder for drugs to diffuse out

Slower diffusion means slower release of drugs

What We're Doing Today

ALGINATE HYDROGEL BEADS

Alginate, a polymer from seaweed, can be cross-linked by calcium via ionic cross-linking. This forms a gel, trapping water inside of hydrogel beads.

- Prepare 1 solution of 1-4% sodium alginate
- Prepare 1 solution of 1% calcium chloride
- Mix dye into alginate solution
- Add alginate solution dropwise into the calcium chloride
- Replace calcium chloride with H_2O
- Take aliquots at 15 min, 30 min, and 1 h to monitor dye release



Q&A